



Why dont we see select context switch

`server_select` disappears after 1,000 connections because it hits the hard system limit of `FD_SETSIZE = 1024`, and prior to 1,000 connections, its data points are directly overlapped by the other server implementations.

- **The 1024 File Descriptor Limit (`FD_SETSIZE`):** Standard C library and Linux implementations of the `select()` system call use fixed-size bitmasks (`fd_set`) bounded by `FD_SETSIZE`, which defaults to 1024. The vertical dotted line at 1K indicates this exact threshold. Attempting to pass file descriptor values ≥ 1024 to `select()` causes buffer overflows or invalid argument errors (`EINVAL`), preventing `server_select` from handling more than 1,024 concurrent connections.
- **Overlapping Data Points ($N \leq 1000$):** For connection counts under 1,024, `server_select` is actually plotted, but its line rests directly beneath `server_poll`, `server_epoll`, and `server_iouring`. At low connection volumes, involuntary context switches across all four I/O multiplexing strategies are virtually identical.